Optimal Algorithms and Inapproximability Results for Every CSP?

by Prasad Raghavendra

Andreas Panayi

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Approximation Algorithms
MSc: A.L.MA. - ΠΜΣ: Α.Λ.ΜΑ.
National Kapodistrian University of Athens - National Technical University of Athens

Main Meta-Theorem (informal)

In this presentation we will see that: assuming the Unique Games Conjecture, for every Constraint Satisfaction Problem:

- The optimal approximation ratio equals the integrality gap of a single natural SDP.
- That ratio is attained by solving the SDP and applying a generic rounding.

Preliminaries

Constraint Satisfaction Problem

A Constraint Satisfaction Problem (CSP) $\Lambda = ([q], \mathbb{P}, k)$ is specified by:

- a *finite domain* $[q] = \{0, 1, \dots, q-1\}$ and
- a set of *predicates* $\mathbb{P}$ (predicate $P : [q]^r \to \{0,1\}, r \leq k$).

The *arity* k of the CSP Λ is the maximum number of inputs to a predicate in $\mathbb{P}$.

Instance of a Constraint Satisfaction Problem

Every instance of the problem $\Lambda = [\mathcal{V}, \mathbb{P}_{\mathcal{V}}]$ consist of:

- a set of *variables* $\mathcal{V}$ and
- a set of *constraints* on them; each constraint consists of a predicate from $\mathbb{P}$ applied to a subset of variables.

The objective is to assign values from $[q]$ to the variables in $\mathcal{V}$ so as to *maximize* the number of satisfied constraints.

Example (CSP): Max-Cut

Problem Definition.

- Domain: $[q] = \{0, 1\}$.
- Predicate set: $\mathbb{P} = \{\mathsf{NEQ}\}$ where $\mathsf{NEQ}(a, b) = 1$ iff $a \neq b$.

- Arity: $k = 2$.

Instance. A graph $G = (V, E)$. Define $\mathcal{V} = \{x_v \mid v \in V\}$ and $\forall\ (u, v) \in E$ include the constraint $(\mathsf{NEQ}, (x_u, x_v)) \in \mathbb{P}_{\mathcal{V}}$.

Objective. Find an assignment $x : \mathcal{V} \to \{0, 1\}$ that maximizes the number of edges (u, v) with $x_u \neq x_v$.

Generalized Constraint Satisfaction Problem

A Generalized CSP $\Lambda = ([q], \mathbb{P}, k)$ is allowed to have a more general predicate function:

$$P : [q]^r \to [-1, 1].$$

Also, every GCSP instance $\Phi = (\mathcal{V}, \mathbb{P}_{\mathcal{V}}, W)$ has additional positive weights $W = \{w_S\}_{S \subseteq \mathcal{V}}$ satisfying $\sum_{S \subseteq \mathcal{V}, |S| \leq k} w_S = 1$. [1]

Objective. Find an assignment to the variables that maximize:

$$\mathbb{E}_{S \in W}[P|_S(y|_S)] = \sum_{S, |S| \leq |k|} w_S P_S(y|_S)$$

[1]By $S \in W$, we denote a set S chosen from the probability distribution W

More Examples

The following problems can also be written as (G)CSP:

- Max-Cut
- Max-k-Cut
- Max Di-Cut
- Max-k-SAT
- Max-k-CSP
- Max-3LIN
- Max-k-Coloring
- Multiway Cut
- Metric Labeling
- Max Acyclic Subgraph
- Label Cover
- Unique Games

and more ...

Conjecture. For every constant $1 > \delta > 0$ there exists a large enough integer R such that the following promise problem is NP-hard.

Instance. A *bipartite* Unique-Games instance

$$\Gamma = (X \cup Y,\ E,\ \Pi = \{\pi_e : [R] \to [R]\}_{e \in E},\ [R]),$$

An assignment of labels $A : X \cup Y \to [R]$ is said to satisfy an edge $e = (v, w)$, if $\pi_{vw}(A(v)) = A(w)$.

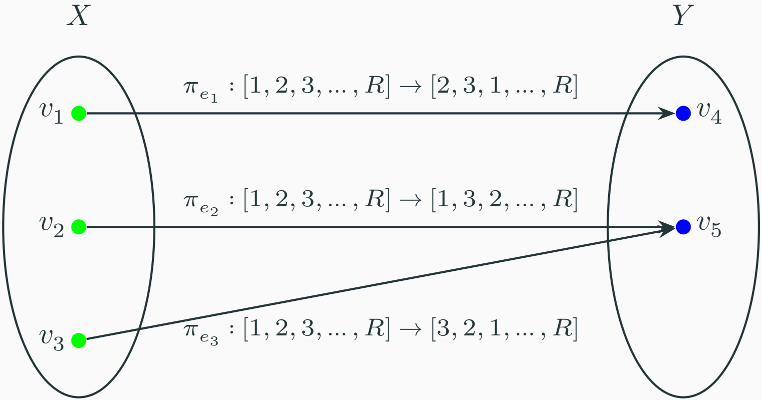
Unique Games Conjecture (2)

Task. Distinguish between the two cases:

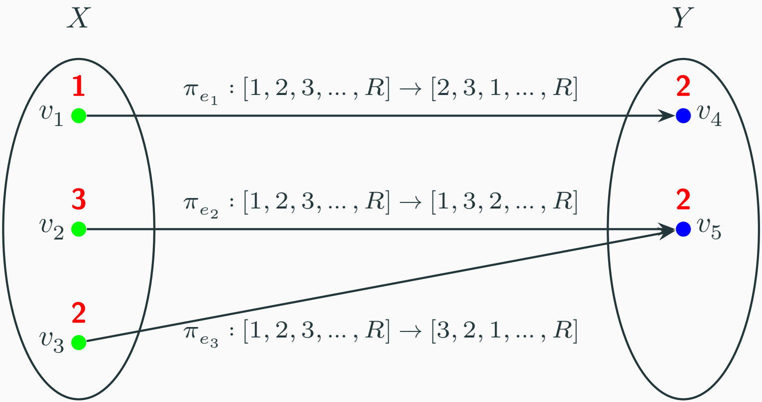
YES: Γ is $(1-\delta)$-satisfiable: $\exists$ assignment $A : X \cup Y \to [R]$ such that for at least $(1-\delta)$-fraction of vertices $v \in X$, *all* edges incident on v are satisfied.

NO: Γ is not δ-satisfiable: every labeling satisfies at most a δ-fraction of the edges of E.

Unique Games Instance: Example



Unique Games Instance: Example



Labeling $A : X \cup Y \to [1, 2, \dots, R]$ where
$A(v_i)$ is equal to the red value over the vertex

Dictator function

A function

$$f : \{0,1\}^R \to \{0,1\}$$

is called a dictator if there exists an index $i \in [R]$ such that

$$f(x_1, \dots, x_R) = x_i.$$

Dictatorship test

A dictatorship test is a randomized test that, given oracle access to

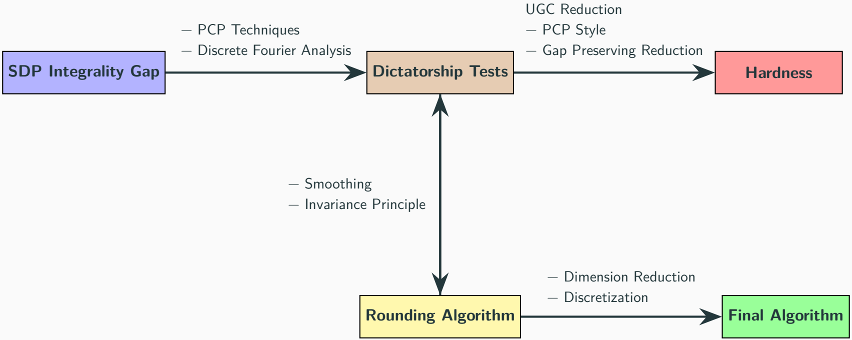
$$f : \{0,1\}^R \to \{0,1\},$$

makes a few queries and distinguishes between:

- (**Completeness**) f is a dictator $\Rightarrow$ the test accepts with high probability;
- (**Soundness**) f is far from every dictator $\Rightarrow$ the test accepts with low probability.

Main Results

Roadmap



— PCP Techniques
— Discrete Fourier Analysis
UGC Reduction
— PCP Style
— Gap Preserving Reduction
SDP Integrality Gap
Dictatorship Tests
Hardness
— Smoothing
— Invariance Principle
— Dimension Reduction
— Discretization
Rounding Algorithm
Final Algorithm

Assume a GCSP $\Lambda = ([q], \mathcal{P}, k)$ and an instance $\Phi = (\mathcal{V},\ \mathbb{P}_{\mathcal{V}},\ W)$.

We introduce the SDP variables

$$\{\, v_{(i,c)} \mid i \in \mathcal{V},\ c \in [q] \,\}, \qquad \{\, X_{(S,\beta)} \mid S \subseteq \mathcal{V},\ \beta \in [q]^S \,\}.$$

Step 1: SDP Relaxation (2)

Maximize $$\sum_{S \in W} w_S \Big(\sum_{\beta \in [q]^S} P_S(\beta(S)) \, X_{(S,\beta)} \Big)$$

Subject to

$$v_{(i,c)} \cdot v_{(i,c')} = 0 \qquad \forall i \in [m],\ c \neq c' \in [q]$$

$$\sum_{c \in [q]} v_{(i,c)} = \mathbf{I} \qquad \forall i \in [m]$$

$$\sum_{\substack{\beta \in [q]^S \\ \beta(s)=c,\ \beta(s')=c'}} X_{(S,\beta)} \ = \ v_{(s,c)} \cdot v_{(s',c')} \qquad \forall S \in W,\ s, s' \in S,\ c, c' \in [q]$$

$$|\mathbf{I}|^2 = 1$$

Step 2: Dictatorship test ($DICT_\Phi(F)$)

Input: SDP solution for Φ, small $\varepsilon > 0$, (depends of oracle F):

1. Sample a constraint $S \in W$.
2. Draw a batch of query points from the SDP-local distribution for S.
3. Apply small independent noise to each query point to obtain $\tilde{Z}_s$.
4. Query the oracle $F : \Omega^R \to \mathrm{Conv}(\Delta_q)$ on each $\tilde{Z}_s$.
5. Return the predicate payoff $P_S(F(\tilde{Z}_{s_1}), \ldots, F(\tilde{Z}_{s_{|S|}}))$.

Step 2: Dictatorship test — completeness

Let $\mathrm{DICT}_\Phi(F)$ be the expected payoff of the test on oracle $F : \Omega^R \to \mathrm{Conv}(\Delta_q)$.

Completeness. If F is a dictator then

$$\mathrm{DICT}_\Phi(F) \ = \ \mathrm{FRAC}(\Phi) \ \pm \ o(1),$$

where the $o(1)$-term vanishes as $\varepsilon \to 0$.

Step 3: Rounding Procedure ($Round_{\mathcal{F}}$)

Input: GCSP instance $\Phi = (\mathcal{V}, \mathbb{P}_{\mathcal{V}}, W)$, SDP solution $\{v_{(i,c)}, X_{(S,\beta)}\}$, $(F : [q]^R \to \text{Conv}(\Delta_q))$

1. Sample R independent Gaussian vectors $\mathbf{g}_1, \dots, \mathbf{g}_R$.
2. For every variable $i \in [m]$:
   (i) Compute the projections $h_{i,c}^{(j)}$ and add noise $n(\varepsilon, \mathbf{g}_j, v_{(i,c)})$:

   $$h_{i,c}^{(j)} = \langle \mathbf{v}_{i,c}, \mathbf{g}_j \rangle + n$$

   (ii) Evaluate the function $\mathcal{F}$ with $h_{i,c}^{(j)}$ as inputs:

   $$\mathbf{p}_i = (p_{(i,0)}, \dots, p_{(i,q-1)}) = \mathcal{F}\left(\{h_{i,c}^{(j)}\}_{c,j}\right)$$

   (iii) Truncate & Normalize to ensure a valid probability distribution:

   $$\tilde{\mathbf{p}}_i \leftarrow \frac{\mathbf{p}_i}{\|\mathbf{p}_i\|_1}$$

   (iv) Set label $x_i = c$ with probability $(\tilde{\mathbf{p}}_i)_c$.

Step 3: Dictatorship test — soundness

Let $F : \Omega^R \to \mathrm{Conv}(\Delta_q)$ and $\mathsf{Round}_F(\Phi)$ be as above.

Soundness (statement). If F is pseudorandom, then

$$\mathrm{DICT}_\Phi(F) \; = \; \mathrm{Round}_F(\Phi) \; \pm \; o(1),$$

and consequently

$$\mathrm{DICT}_\Phi(F) \; \leq \; \mathrm{INT}(\Phi) \; + \; o(1),$$

where the $o(1)$-errors vanish as $\varepsilon \to 0$.

Step 4: Hardness Reduction (UGC $\leq$ GCSP$_\Lambda$)

- Fix a GCSP problem Λ and a base instance $\Phi \in \Lambda$.
- Construct the *dictatorship test* DICT$_\Phi$ from the SDP solution of Φ.
- Given a Unique Games instance $\Gamma = (X \cup Y, E, \{\pi_{vw}\})$:
  - For each $w \in Y$, introduce long-code variables $\mathcal{F}^w : [q]^R \to \Delta$.
  - For each $v \in X$, define induced oracle

$$\mathcal{F}^v(z) = \mathbb{E}_{w \in N(v)}[\mathcal{F}^w(\pi_{vw}(z))].$$

- Run DICT$_\Phi$ on $\mathcal{F}^v$ to obtain queries and payoffs.
- Simulate each oracle query via neighbor queries to $\mathcal{F}^w$.
- Translate each test execution into GCSP constraints over variables

$$Y \times [q]^R.$$

Step 4: Completeness and Soundness of the Reduction

- **Completeness:** If Γ has a near-perfect labeling, by setting each $\mathscr{F}^w$ to the corresponding dictator of that label, yields $\Phi(\Gamma)$ of value $\approx$ SDPval(Φ).
- **Soundness:** If the Unique Games instance has no solution, no consistent we cannot form consistent Dictators. Hence, using the Invariance Principle, the CSP value drops to $\approx$INT(Φ)

Step 5: Algorithm

- **Finite cover:** Construct a *finite* family $\mathcal{F}_\kappa = \{F_1, \dots, F_l\}$ for the space of functions $F : [q]^R \to \text{Conv}(\Delta_q)$ such that for any function F, there exists $F_i \in \mathcal{F}_\kappa$ with $\|F - F_i\|_\infty \leq \kappa$.
- **Enumerate & round:** For every function $F \in \mathcal{F}_\kappa$, run the rounding scheme $\text{Round}_F(\Phi)$ and output the best assignment found.
- **Guarantee:** The algorithm outputs a solution with optimal value assuming the UGC.

Unconditional result (2-CSPs)

For Boolean 2-CSPs, the canonical SDP together with the Round procedure achieves the integrality-gap optimal approximation ratio unconditionally (no UGC required).

This holds because Khot–Vishnoi give explicit integrality-gap instances for the 2-CSP SDP, and the Round analysis matches those gaps directly.

Further Work and Remarks

Raghavendra–Steurer (FOCS 2009): Algorithmic Follow-Up

- Consider the *basic SDP relaxation* for a fixed CSP problem.
- For every SDP solution with value ρ, there is a *polynomial-time rounding algorithm* producing an integral solution of value at least

$$\mathrm{Gap}_{\mathsf{SDP}}(\rho) - \varepsilon.$$

- The rounding is *universal*: the same algorithm applies to all CSPs (dependence only on arity, alphabet size, and ε).

Consequently, the integrality gap of the basic SDP is an algorithmically achievable approximation ratio.

Final Remarks

- **Meta Algorithm:** A single "master" SDP and generic rounding strategy.
- **Automated Hardness:** The SDP integrality gap directly identifies the optimal approximation lower bound.
- **Pioneer Techniques:** The proof introduced groundbreaking methods, most notably the *Invariance Principle*.

References

References

[Rag08] Prasad Raghavendra. **"Optimal Algorithms and Inapproximability Results for Every CSP?"** In: *Proceedings of the 40th Annual ACM Symposium on Theory of Computing*. STOC. ACM, 2008, pp. 245–254. DOI: 10.1145/1374376.1374414.

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Thank You ☺